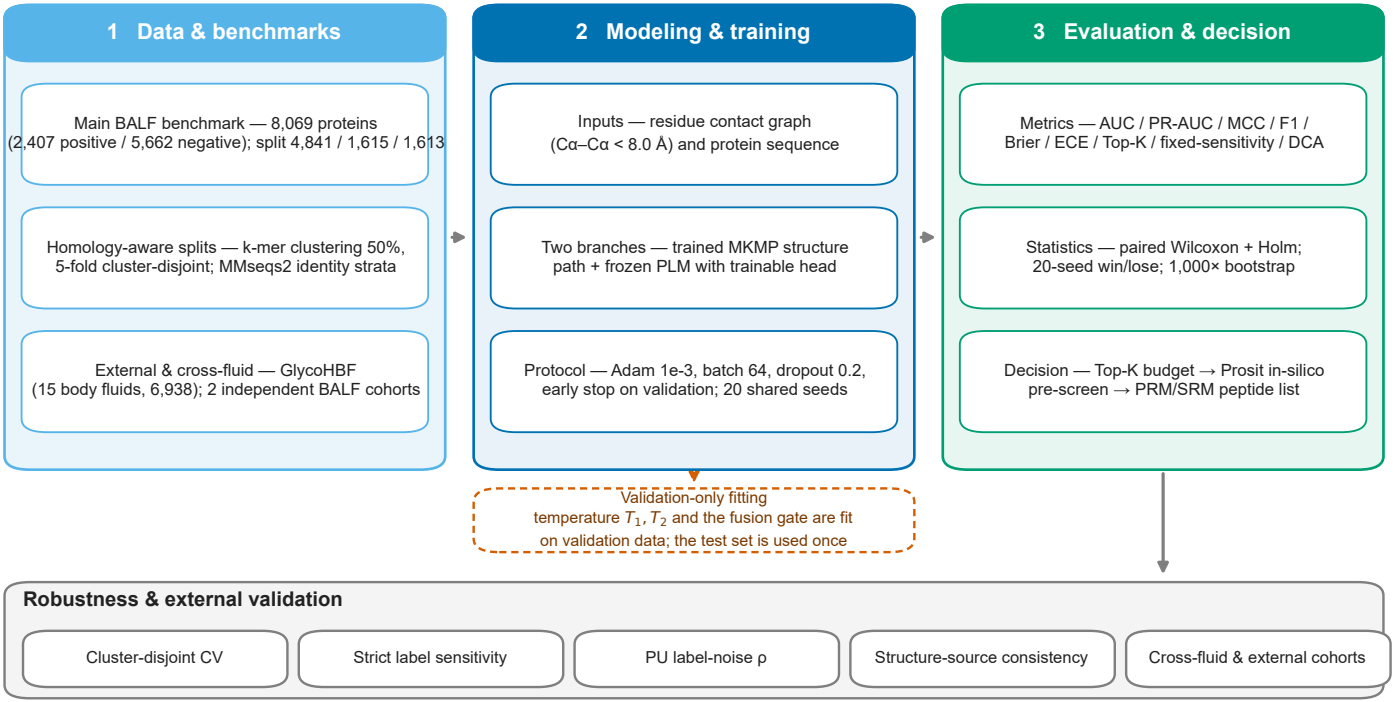

1 Data & benchmarks
Main BALF benchmark — 8,069 proteins (2,407 positive / 5,662 negative); split 4,841 / 1,615 / 1,613
Homology-aware splits — k-mer clustering 50%, 5-fold cluster-disjoint; MMseqs2 identity strata
External & cross-fluid — GlycoHBF (15 body fluids, 6,938); 2 independent BALF cohorts
2 Modeling & training
Inputs — residue contact graph (Cα–Cα < 8.0 Å) and protein sequence
Two branches — trained MKMP structure path + frozen PLM with trainable head
Protocol — Adam 1e-3, batch 64, dropout 0.2, early stop on validation; 20 shared seeds
Validation-only fitting
temperature T_1, T_2 and the fusion gate are fit on validation data; the test set is used once
3 Evaluation & decision
Metrics — AUC / PR-AUC / MCC / F1 / Brier / ECE / Top-K / fixed-sensitivity / DCA
Statistics — paired Wilcoxon + Holm; 20-seed win/lose; 1,000× bootstrap
Decision — Top-K budget → Prosit in-silico pre-screen → PRM/SRM peptide list
Robustness & external validation
Cluster-disjoint CV
Strict label sensitivity
PU label-noise ρ
Structure-source consistency
Cross-fluid & external cohorts